

A Structural VAR of the Global Oil Market: A Replication on Public Data

Replication of Bank of Canada Staff Analytical Note 2019-17

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Abstract

This note reproduces the structural vector autoregression (SVAR) of the global oil market in Ellwanger (2019), Bank of Canada Staff Analytical Note 2019-17, using publicly available proxies. The model is a four-variable VAR identified through the note’s elasticity impact matrix, with the short-run price elasticity of oil supply fixed as the single identifying restriction. We recover the four structural shocks—supply, oil-market-specific demand, storage demand, and economic growth—and report impulse responses, a forecast-error variance decomposition, a historical decomposition of the real oil price, and two policy scenarios (an OPEC+ supply cut and a global-growth revision). The implementation is a faithful re-creation of the model’s structure and qualitative story; because it uses public proxies rather than the note’s proprietary data, the numerical values differ by construction.

1 Method

We follow the structural model of Ellwanger (2019). Let $y_t = (q_t^{\text{prod}}, q_t^{\text{cons}}, p_t, a_t)'$ collect world oil production, world oil consumption, the real price of oil, and a measure of global economic activity, the first three in logs and activity in levels. The reduced form is a VAR of order p with a constant and seasonal dummies,

$$y_t = c + \sum_{s=1}^p A_s y_{t-s} + \Phi d_t + u_t, \quad \mathbb{E}[u_t u_t'] = \Sigma, \quad (1)$$

estimated by OLS. In the quarterly baseline $p = 4$ with three seasonal dummies; d_t additionally holds impulse dummies for the extreme 2020 (COVID-19) quarters so that those observations do not distort Σ or the identification. The largest companion eigenvalue is close to but below one, reflecting the high persistence of the underlying series.

Data. We use public proxies over 1997Q1–2024Q4 (about 108 usable quarters after lags): the real oil price is Brent (FRED POILBREUSD) deflated by US CPI (CPIAUCSL); production and consumption are world “petroleum and other liquids” from the EIA Short-Term Energy Outlook (PAPR_WORLD, PATC_WORLD); and activity is Kilian’s index of global real economic activity (IGREA; Kilian, 2009)—a percentage deviation from trend, rescaled by 1/100, rather than log real GDP, so activity responses are in index units and their magnitude is not directly comparable to the note (setting `cfg.activity='gdp_proxy'` switches to a log activity index for percentage-based responses). These stand in for the note’s proprietary IEA quantities and Bank of Canada activity measure, so the model structure is reproduced but the numbers differ by construction.

The reduced-form residuals u_t are linear combinations of four mutually uncorrelated structural shocks ε_t ,

$$u_t = B \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \varepsilon_t'] = D = \text{diag}(\sigma_1^2, \dots, \sigma_4^2). \quad (2)$$

Identification consists of choosing the structural impact matrix B ; the impulse responses, variance decomposition, historical decomposition, and scenarios all follow from B , D , and the estimated VAR coefficients.

2 Identification

The four structural shocks are a *flow supply* shock (unexpected shifts in oil production), an *oil-market-specific demand* shock (demand shifts not driven by the business cycle, e.g. precautionary or convenience-yield demand for oil), a *storage / speculative demand* shock (demand for above-ground inventories, reflecting expectations about future market tightness), and an *economic growth* shock (innovations to global activity that raise oil demand through income).

Rather than a recursive (Cholesky) ordering, identification uses the economic impact matrix of Ellwanger (2019), parameterised by seven structural elasticities $\theta = (\eta_S, \eta_D, \eta_I, \gamma_S, \gamma_D, \gamma_G, \eta_G)$. Writing the mapping (2) in full makes the identifying structure explicit: each reduced-form residual (a variable's one-step forecast error) is a combination of the four structural shocks, with the columns of $B(\theta)$ giving the contemporaneous impact of each shock,

$$\underbrace{\begin{pmatrix} u_t^{\text{prod}} \\ u_t^{\text{cons}} \\ u_t^p \\ u_t^a \end{pmatrix}}_{u_t} = \underbrace{\begin{pmatrix} 1 & \eta_S \gamma_D & \eta_S & \eta_S(\eta_I \gamma_D + \gamma_G) \\ \eta_D \gamma_S & 1 & \eta_D & \eta_I \\ \gamma_S & \gamma_D & 1 & \eta_I \gamma_D + \gamma_G \\ \eta_G & 0 & 0 & 1 \end{pmatrix}}_{B(\theta)} \underbrace{\begin{pmatrix} \varepsilon_t^{\text{sup}} \\ \varepsilon_t^{\text{oil}} \\ \varepsilon_t^{\text{sto}} \\ \varepsilon_t^{\text{gro}} \end{pmatrix}}_{\varepsilon_t}, \quad (3)$$

where the residual vector orders production, consumption, real price and activity, and the shock vector orders the supply, oil-specific demand, storage-demand and growth shocks. The unit diagonal is a normalisation, so each shock has a unit own-impact and its scale is carried by D ; the two zeros in the last row encode the exclusion restriction that oil-market shocks (oil-specific demand and storage) do not move global activity within the period.

Obtaining the entries. The elements of $B(\theta)$ are not free: they are the contemporaneous impact multipliers implied by the model's behavioural equations. Within the period, oil production adjusts only along a short-run supply curve, oil consumption along an income-dependent demand curve, and global activity is predetermined with respect to the two oil-market demand shocks,

$$u_t^{\text{prod}} = \eta_S u_t^p + \varepsilon_t^{\text{sup}}, \quad (4)$$

$$u_t^{\text{cons}} = \eta_D u_t^p + \eta_I u_t^a + \varepsilon_t^{\text{oil}}, \quad (5)$$

$$u_t^a = \eta_G \varepsilon_t^{\text{sup}} + \varepsilon_t^{\text{gro}}, \quad (6)$$

while the real price clears the market subject to storage arbitrage, responding to the supply shock (γ_S), to oil-specific demand together with the income-driven demand it induces (γ_D), to the storage shock (unit impact), and to the expectations component of the growth shock (γ_G). Solving this simultaneous system for the innovations in terms of the shocks, and normalising each shock to a unit impact on its own variable (the unit diagonal), yields $B(\theta)$ in (3). Two readings illustrate the construction. First, because production moves only through the supply curve (4), the production row equals η_S times the price row across the demand, storage and growth columns: an oil-specific demand shock raises the price by γ_D and hence production by $\eta_S \gamma_D$. Second, the growth-shock

price impact $\eta_I \gamma_D + \gamma_G$ (entry (3, 4)) is the sum of an income channel—higher activity raises oil demand through η_I , which raises the price through γ_D —and a direct expectations channel γ_G .

Table 1 lists the role, expected sign and status of each parameter.

Table 1: Structural parameters in $B(\theta)$.

Parameter	Interpretation	Expected sign	Status
η_S	short-run price elasticity of oil supply	+ (small)	fixed at 0.03
η_D	short-run price elasticity of oil demand	−	estimated
η_I	income (activity) elasticity of demand	+	estimated
γ_S	price impact of a supply shock	−	estimated
γ_D	price impact of an oil-specific demand shock	+	estimated
γ_G	expectations component of the growth-shock price impact	+	estimated
η_G	contemporaneous activity response to a supply shock	+	estimated

Because supply is very inelastic in the short run, the price elasticity of supply η_S is *fixed* as the single identifying restriction. The remaining parameters and the shock variances are then pinned down by matching the model-implied and sample residual covariances. Substituting (3) into the residual covariance gives the moment conditions

$$\Sigma = B(\theta) D B(\theta)' = \sum_{j=1}^4 \sigma_j^2 b_j(\theta) b_j(\theta)', \quad (7)$$

where $b_j(\theta)$ is the j -th column of $B(\theta)$: the residual covariance decomposes into a sum of rank-one contributions, one per structural shock, each the outer product of that shock’s impact vector scaled by its variance. The symmetric 4×4 matrix Σ has ten distinct elements, matched to ten unknowns—the six free elasticities (all of θ except the fixed η_S) and the four shock variances σ_j^2 —so the system is exactly identified given η_S . It is solved numerically by a multi-start optimiser. We set $\eta_S = 0.03$: on the 1997–2024 public sample this value gives the best covariance match and the most nearly orthogonal recovered shocks, while remaining inside the range of short-run supply elasticities in the literature. A sensitivity table over a grid of η_S is produced at run time.

Two caveats attach to the identification on public data. First, the parameters η_I , γ_G , and η_G are only weakly identified by this covariance and are driven toward zero; they can be pinned to zero with no change to the fit or the other elasticities. Second, imposing *all* of the note’s reported elasticities on the public covariance is degenerate (a shock variance collapses to zero), because those values come from an informative-prior / external-elasticity identification on the note’s proprietary data rather than from matching this sample’s covariance.

3 Results

Table 2 reports the identified elasticities and shock standard deviations. The signs are as expected—demand slopes down ($\eta_D < 0$), a supply shock lowers the price impact channel ($\gamma_S < 0$), and an oil-specific demand shock raises it ($\gamma_D > 0$). Consistent with the weak-identification caveat in Section 2, the two weakly identified parameters γ_G and η_G sit at zero. The covariance match is close but not exact, as expected on public data.

Table 2: Identified structural parameters (η_S fixed at 0.03; sample 1997Q1–2024Q4).

Parameter	Estimate
η_S supply price elasticity (fixed)	0.030
η_D demand price elasticity	−0.105
η_I income elasticity of demand	0.018
γ_S price impact, supply shock	−4.599
γ_D price impact, oil-demand shock	9.449
γ_G expectations/growth price impact	0.000
η_G activity response to supply shock	0.000
Shock SDs ($\sigma_{\text{sup}}, \sigma_{\text{oil}}, \sigma_{\text{sto}}, \sigma_{\text{gro}}$)	(0.0086, 0.0092, 0.0653, 0.3129)

Covariance-match fit 8.4×10^{-4} (0 = exact); $\max |\Sigma - BDB'| = 1.8 \times 10^{-4}$; solver `lsqnonlin` (refined). Shocks ordered supply, oil-market demand, storage demand, growth.

Impulse responses. Figure 1 reports responses to one-standard-deviation structural shocks over 20 quarters, with 68% bootstrap bands (moving-block residual bootstrap, block length $\approx T^{1/3}$). A supply shock lowers production and raises the real price; an oil-market demand shock and a growth shock both raise the real price, the former on impact and the latter more persistently; the storage-demand shock reallocates between current production and consumption. The plotted central line is the bootstrap median, so it always lies inside the band.

Variance decomposition. Figure 2 shows the forecast-error variance decomposition. For the real oil price, oil-market demand dominates at short horizons while the supply and growth shocks account for a growing share as the horizon lengthens—the population counterpart to the in-sample shares in the historical decomposition.

Historical decomposition. Figure 3 decomposes the real oil price (relative to its deterministic-plus-initial-conditions baseline) into the cumulative contribution of each shock, quarter by quarter. The solid black line is the total (data minus baseline); the stacked bars are the four shocks and sum to it. The reconstruction error is numerically zero. The plotting window can be trimmed (here it starts in 2021) while estimation still uses the full sample; the COVID quarters carry impulse dummies, so their bars are netted out.

4 Scenarios

Both scenarios are conditional forecasts: a target path is imposed on one variable and delivered by allowing only the associated structural shock to move (a minimum-variance / Waggoner–Zha conditioning), with all other shocks left at their unconditional values. The horizon is $H = 8$ quarters. Figure 4 shows the resulting real-oil-price paths, in per cent and US dollars (converted at the most recent Brent level), with 68% bootstrap bands. Both magnitudes are illustrative and set in the configuration (`supOpt.cutMbd`, `imfOpt.endRev`).

Supply: an OPEC+ production cut. World production is conditioned onto a lower path via the *supply* shock: a cut of 1.0 Mb/d, phased in linearly over the first 2 quarters and then held for the remaining horizon. A supply cut raises the real price; the response scales with the identified supply price impact γ_S .

Demand: a global-growth revision. Activity is conditioned onto a downward-revised path via the *economic-growth* shock, ramped linearly to -0.3 activity-innovation standard deviations by the end of the horizon. A downward growth revision lowers the real price. Fixing both scenario magnitudes ex ante (from the estimated VAR) ensures the point estimate and every bootstrap replication impose the same revision, so the bands reflect parameter uncertainty rather than collapsing.

Figure 5 additionally reports a conditional-forecast fan: conditioning on anticipated higher and lower activity paths (fed through the growth shock) and reading off the implied paths for production, consumption, net stock build, the real price, and activity.

5 Conclusion

The exercise reproduces the structure and qualitative economics of Ellwanger (2019) on public proxies: an inelastic-supply, elasticity-based identification of four oil-market shocks, with sensible impulse responses, a demand-then-supply/growth pattern in the variance decomposition, an interpretable historical decomposition of the real price, and two transparent policy scenarios. Because the data are public proxies and the identification matches this sample’s covariance rather than importing external elasticities, the numbers differ from the note; the closest numerical replication would use the note’s own production, consumption, and activity series together with its calibrated identification.

References

- Ellwanger, R. (2019). *A Structural Model of the Global Oil Market*. Bank of Canada Staff Analytical Note 2019-17. <https://www.bankofcanada.ca/2019/06/staff-analytical-note-2019-17/>
- Kilian, L. (2009). Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market. *American Economic Review*, 99(3), 1053–1069.

A Figures

IRFs to one-SD structural shocks (per cent; line = bootstrap median, shaded = 68% band)

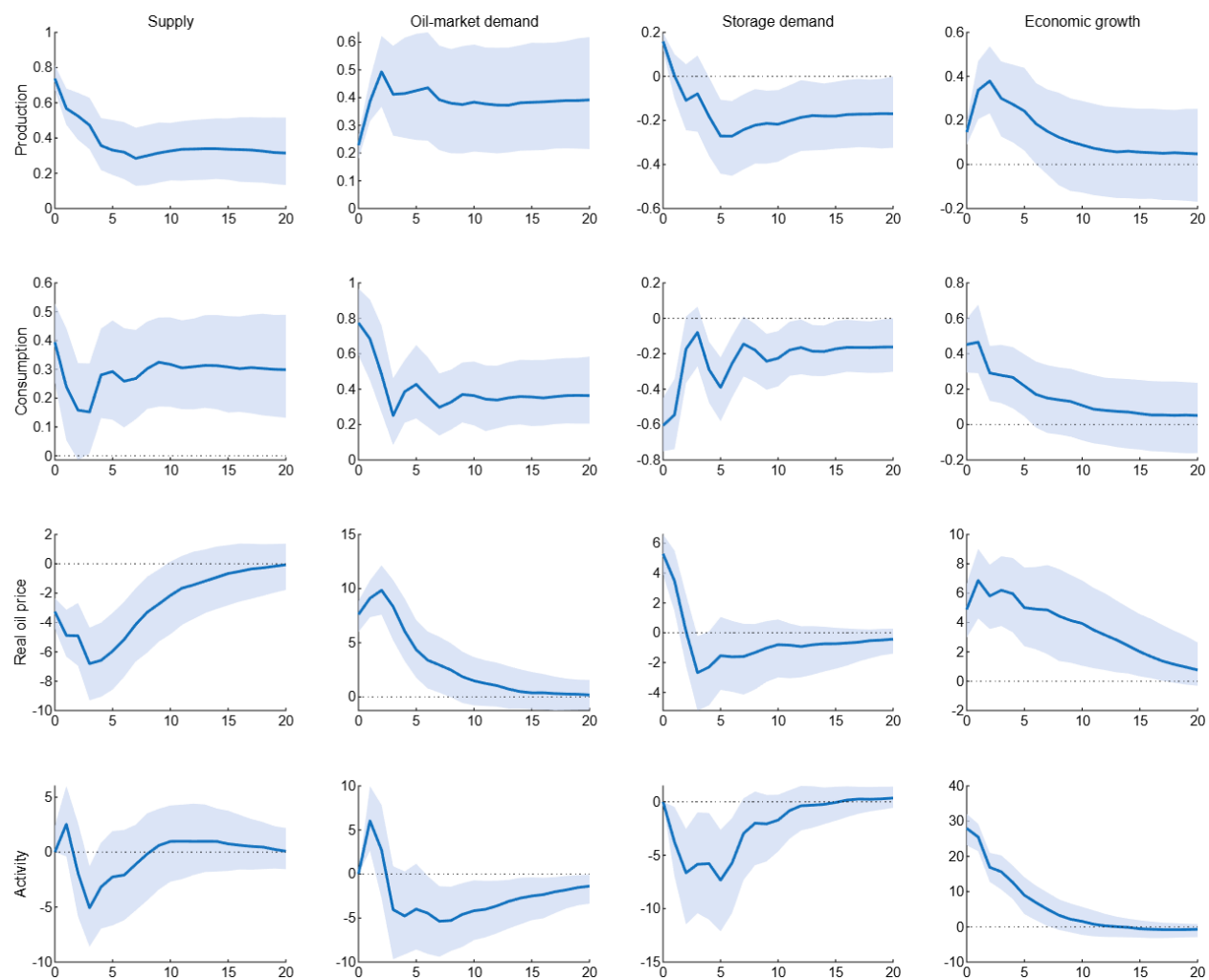


Figure 1: Impulse responses to one-standard-deviation structural shocks (rows: variables; columns: shocks). Central line is the bootstrap median; shaded area is the 68% band. Referenced in Section 3.

Forecast-error variance decomposition (share of each shock by horizon)

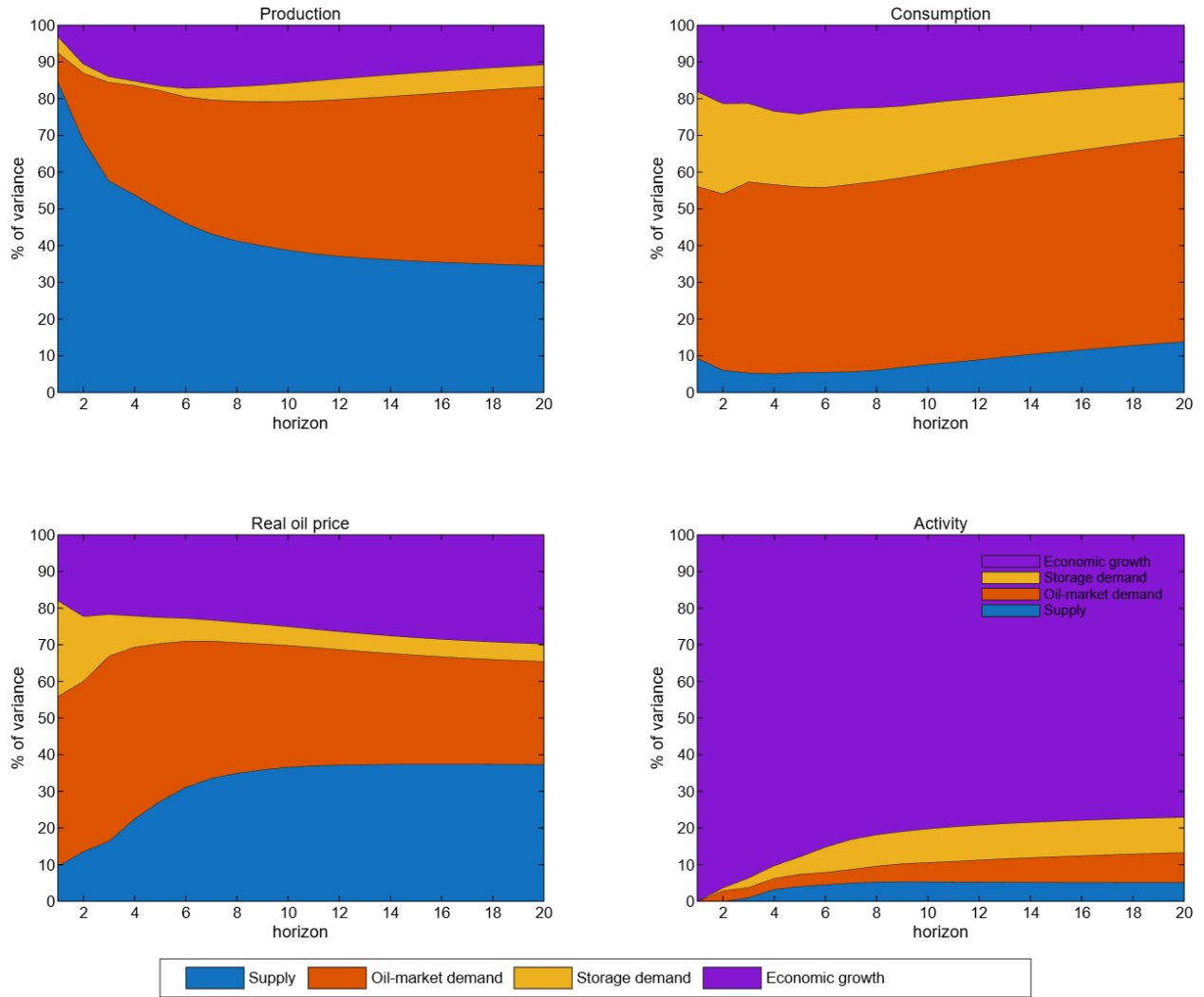


Figure 2: Forecast-error variance decomposition. Each panel is one variable; the stacked areas are the shares of the four shocks by horizon. Referenced in Section 3.

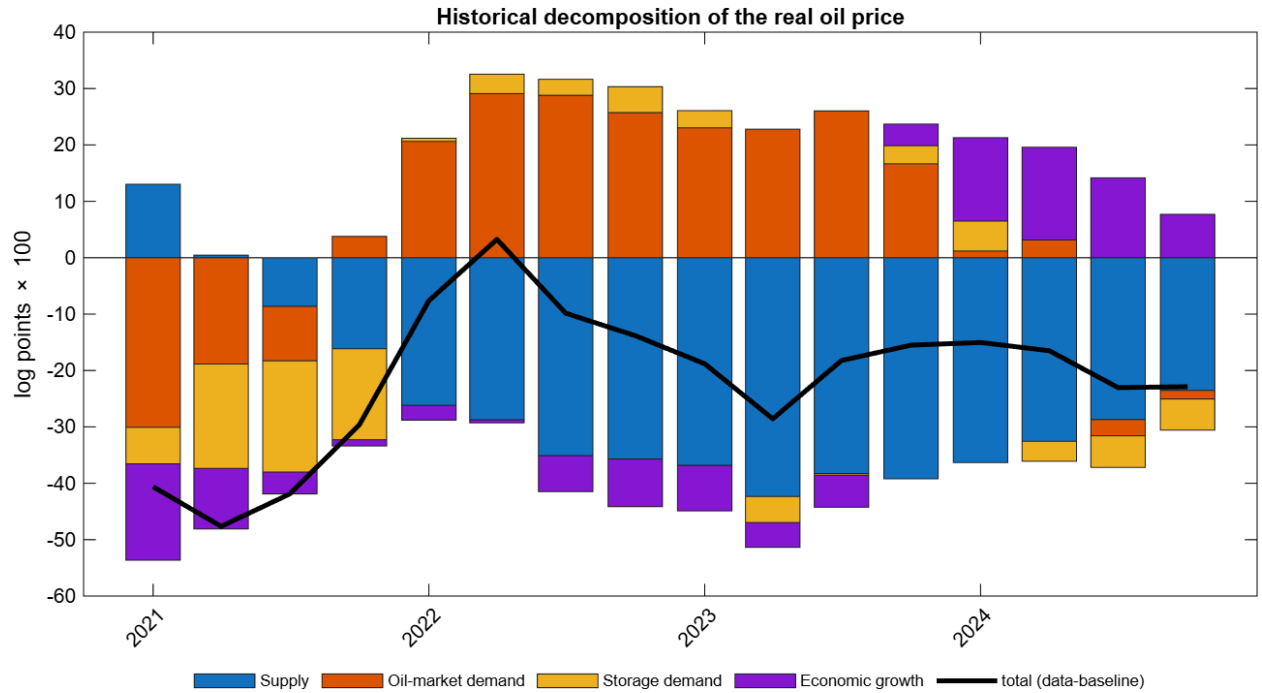


Figure 3: Historical decomposition of the real oil price. Stacked bars are the four shocks' cumulative contributions; the solid line is the total (data minus baseline). Referenced in Section 3.

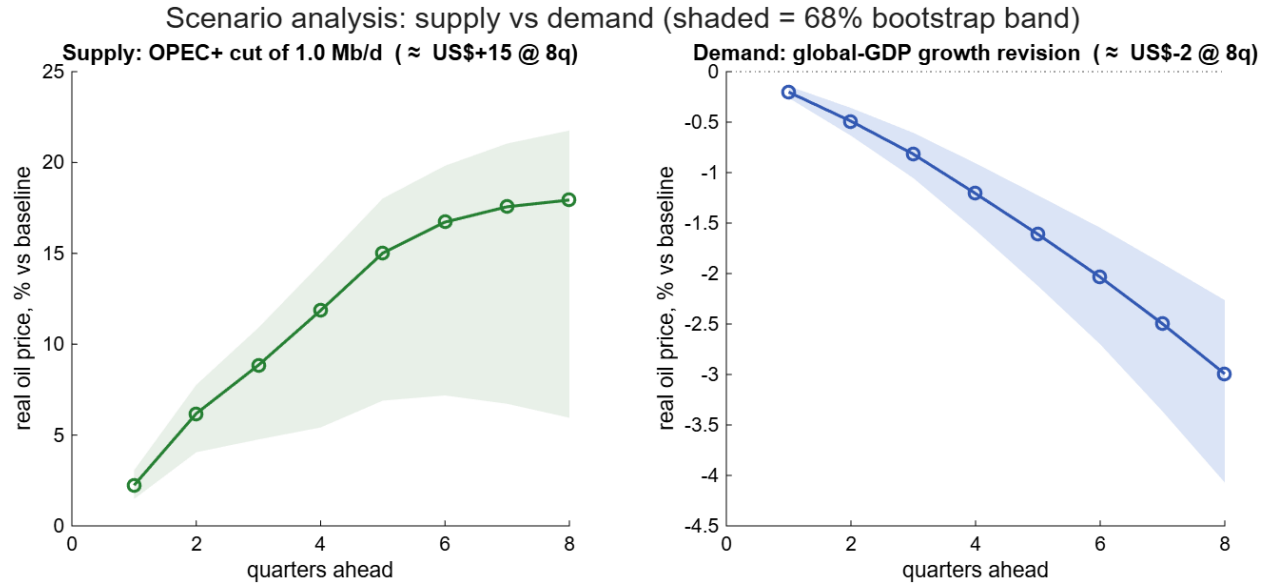


Figure 4: Scenario analysis. Left: OPEC+ supply cut of 1.0 Mb/d (supply shock). Right: global-growth revision of -0.3 activity-innovation SD (growth shock). Real-oil-price response in per cent, with 68% bootstrap bands; dollar figures in the panel titles. Referenced in Section 4.

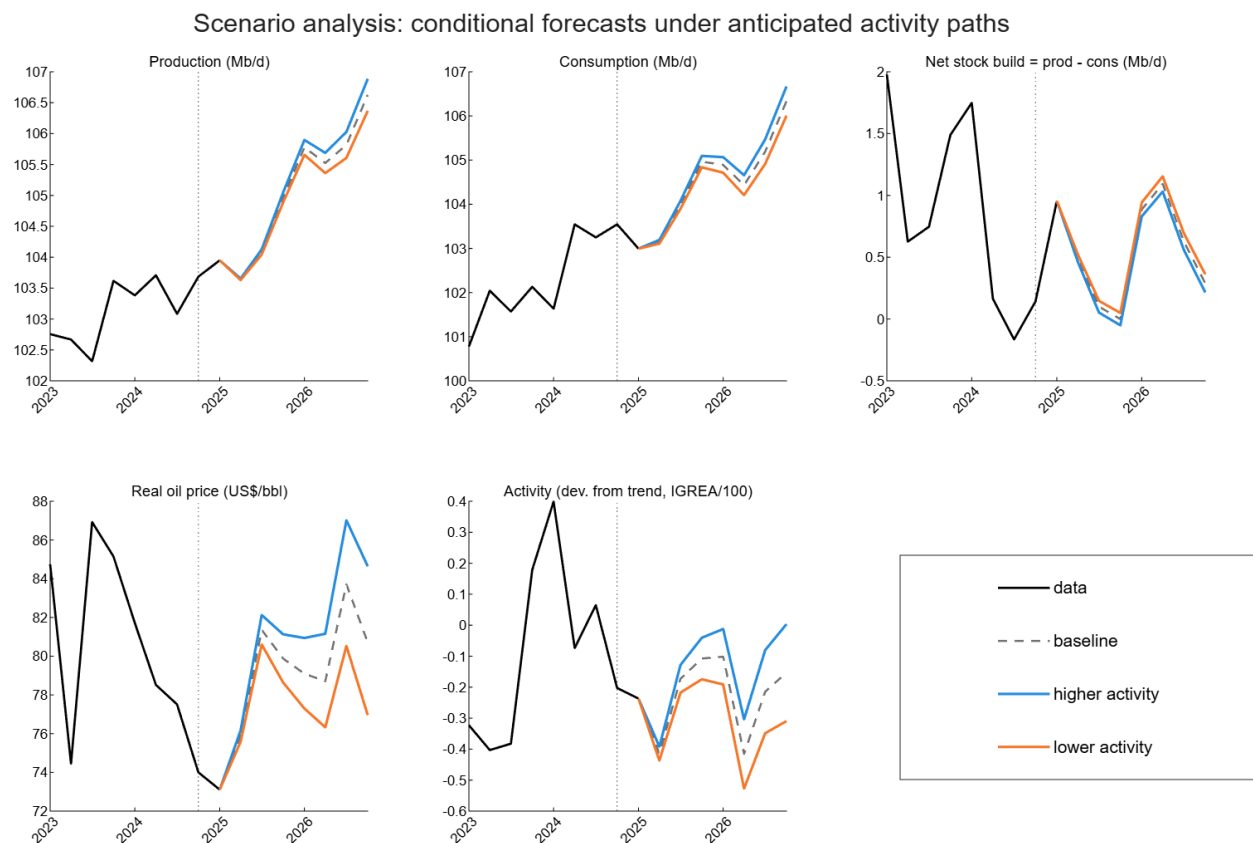


Figure 5: Conditional forecasts under anticipated activity paths (higher vs lower activity), showing the implied paths of production, consumption, net stock build, the real oil price, and activity. Referenced in Section 4.